

Dr. rer. nat. Basavaraj Hampiholi

AI Researcher & Data Scientist | Nürnberg, Germany | basavaraj_hampiholi@outlook.com

Computer vision researcher and data scientist with a PhD in Artificial Intelligence. Research spans multimodal action recognition, video understanding, clinical AI, federated learning and scalable data infrastructure.

PROFESSIONAL EXPERIENCE

Scientific Research Associate | Data Scientist

Mar 2023 – Present

Ulm University, Germany

- Designed and implemented an end-to-end FAIR-aligned clinical data lake on Kubernetes, integrating MinIO, ETL, analytics and machine learning tools.
- Applied federated learning through PrivateAIM to distributed clinical data, including an end-to-end pancreatic cancer use case.
- Collaborated with technical and domain stakeholders on reproducible clinical analytics and ML infrastructure.

PhD Student | Deep Learning & Computer Vision

Dec 2018 – Feb 2022

BMW Car IT GmbH, Ulm, Germany

- Developed efficient architectures for fine-grained temporal action segmentation in untrimmed video.
- Proposed transformer-based multimodal fusion for activity recognition and representation learning.
- Contributed research in multimodal learning and language-guided video understanding, published in peer-reviewed venues.

Machine Learning Intern | Computer Vision

Apr – Sep 2018

NEC Laboratories Europe, Heidelberg, Germany

- Developed deep learning methods for mitotic cell-division phase detection and tracking in microscopy.
- Built video analytics for urban monitoring, including illegal-dumping detection.

Data Engineer | Manufacturing Analytics

Dec 2011 – Nov 2014

Tata Consultancy Services, Pune, India

- Built ingestion and preprocessing pipelines for industrial sensor, historian, planning and quality-system data.
- Developed SQL- and R-based workflows for production reporting and predictive quality analytics.

EDUCATION

PhD, Artificial Intelligence — Ulm University, Germany | 2018–2026 | magna cum laude

Thesis: Learning-based Multimodal Action Recognition

MSc, Computer Science (Intelligent Systems) — Technical University Kaiserslautern, Germany | 2015–2018 | Grade 1.9

Thesis: Learning 3D Shapes as Multi-Layered Height Maps Using 2D CNNs

BEng, Computer Science — Visvesvaraya Technological University, India | 2007–2011 | CGPA 8.46/10

SELECTED PUBLICATIONS

- Hu, X., **Hampiholi, B.**, Neumann, H., Lang, J. (2024). Temporal Context Enhanced Referring Video Object Segmentation. *WACV*.
- **Hampiholi, B.**, Jarvers, C., Mader, W., Neumann, H. (2023). Convolutional Transformer Fusion Blocks for Multi-Modal Gesture Recognition. *IEEE Access*.
- **Hampiholi, B.**, Jarvers, C., Mader, W., Neumann, H. (2020). Depthwise Separable Temporal Convolutional Network for Action Segmentation. *3DV*.

- Sarkar, K., **Hampiholi, B.**, Varanasi, K., Stricker, D. (2018). Learning 3D Shapes as Multi-Layered Height Maps Using 2D Convolutional Networks. *ECCV*.

TECHNICAL SKILLS

Programming & vision: Python, NumPy, SciPy, Pandas, scikit-image, OpenCV, SQL, PySpark, Tesseract

ML & AI: PyTorch, scikit-learn, Hugging Face, LangChain, SHAP, LIME

Data & infrastructure: MLflow, Apache Iceberg, MinIO, Docker, Kubernetes

Systems & languages: Linux, Windows, macOS, ROS; English (proficient), German (B1)

AWARDS

Star of the Month and On the Spot awards (TCS); Best Idea award (EESTEC Kaiserslautern).